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|---------------|-----------------------------------------------------------|
| <b>Title</b>  | <b>Kryptos K4. Evidence for a 44 Character Plain Text</b> |
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| <b>Date</b>   | <b>10<sup>th</sup> October 2010</b>                       |

## Introduction

The following document sets out a series of what are assumed to be novel discoveries in relation to the remaining unsolved cipher texts embedded in Jim Sanborn's Kryptos and Antipodes sculptures.

Whereas this does not purport to be a complete solution it does offer a candidate for the decryption of K4 per se and is hoped in time to mark a significant waypoint on the road to the eventual discovery of the outstanding plain text. The various statements of what are thought to be novel discoveries have been grouped and presented according to their viability.

The readership is aimed primarily at those in the Kryptos K4 community who are already conversant with the cipher and plain texts of sections K1-K3 as well as the associated tools and techniques. Such readers may wish to skip the background notes and proceed directly to the later paragraphs.

As with sections K1, K2 and K3 anyone wishing to repeat these results for themselves needs only the decryption tableau, a pen and some paper.

I'm indebted to all the hard work undertaken in both the recognized forums and micro sites that have sprung up along the way in support of the commonly shared goal and hope that in time this work finds a place in the canon of standard references used to offer a complete decryption. Should any aspects of the work infringe anyone's prior art then my sincere apologies are offered as I have been as exhaustive as possible in searching the public domain for blogs or micro sites venturing a similar solution.

This introduction would be incomplete without a sincere expression of gratitude to Jim Sanborn, Ed Scheidt and their Federal patrons for seeding the world with such a constructive curiosity.

Lastly I'd like to thank Elonka Dunin for taking the time out to cast an eye over the first draft and offer constructive feedback. Anyone unfamiliar with Kryptos may also want to visit her Kryptos micro site at <http://www.elonka.com/kryptos> before proceeding any further with this document.

## Background

*The following information on the background of the Kryptos sculpture was gathered from a range of internet sources.*

Kryptos was dedicated on November the 3<sup>rd</sup> 1990 on the grounds of the CIA headquarters in Langley Virginia. The main body of the sculpture is a worked copper scroll by Artist Jim Sanborn who was commissioned by the CIA to create the installation for the courtyard of their headquarters. He was additionally loaned the services of a retiring CIA employee by the name of Ed Scheidt who was on hand to provide assistance with the cryptographic elements of the piece.

Since 1990 there have been three public acknowledgments in relation to the decryption of parts K1, K2 and K3.

*A civilian computer scientist named Jim Gillogly went public with a decryption of parts K1, K2 and K3 in 1999 having used computational analysis to arrive at the plain texts.*

*After this announcement it was revealed that a CIA employee named David Stein was recognised within the CIA as having cracked parts K1, K2 and K3 using a pencil and paper technique in 1998.*

*It was later announced in 2005 that an NSA team led by Ken Miller, Dennis McDaniels and two other unnamed individuals had deciphered parts K1, K2 & K3 in late 1992 using computational analysis.*

In the years since Gillogly's announcement a mystique has continued to grow around the unsolved K4 cipher text. This is partly due to its resistance to yield its final secrets, but also the mass of interest aroused worldwide by Dan Brown's code sleuthing novels and the growing pervasiveness of cryptography in modern life. As a result there has been a mass of internet based collaboration in the public domain ever since, whereby common interest groups and individuals have persistently poured over the clues left in the plain texts of parts K1, K2 and K3 in search of possible routes into the plain text of K4.

For anyone not familiar with plain texts of K1, K2 and K3 they read as follows.

### ***K 1 (this was encrypted using the Keyword PALIMPSEST)***

BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT LIES THE NUANCE OF IQLUSION

***K2 (this was encrypted using the Keyword ABSCISSA)*** *Note: This was publicly corrected by Jim Sanborn on April 19th, 2006 who intervened to point out the IDBYROWS should read XLAYERTWO*

IT WAS TOTALLY INVISIBLE HOWS THAT POSSIBLE ? THEY USED THE EARTHS MAGNETIC FIELD X THE INFORMATION WAS GATHERED AND TRANSMITTED UNDERGRUUND TO AN UNKNOWN LOCATION X DOES LANGLEY KNOW ABOUT THIS ? THEY SHOULD ITS BURIED OUT THERE SOMEWHERE X WHO KNOWS THE EXACT LOCATION ? ONLY WW THIS WAS HIS LAST MESSAGE X THIRTY EIGHT DEGREES FIFTY SEVEN MINUTES SIX POINT FIVE SECONDS NORTH SEVENTY SEVEN DEGREES EIGHT MINUTES FORTY FOUR SECONDS WEST X LAYER TWO

*K 3 (this section was not encrypted using the Kryptos Tableau, but instead made use of a 192 character skip sequence to encrypt the plain text)*

SLOWLY DESPARATLY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED WITH TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER AND THEN WIDENING THE HOLE A LITTLE I INSERTED THE CANDLE AND PEERED IN THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER BUT PRESENTLY DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST X CAN YOU SEE ANYTHING Q

*Complete Cipher Text and Decryption Tableau as it appears on the sculpture. This has been lifted directly from Elonka Dunin's Kryptos site at <http://elonka.com/kryptos/transcript.html>.*

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>EMUFPHZLRFAXYUSDJKZLDKRNSHGNFIVJ<br/> YQTQUXQBQVYUULLTREVJYQTMKYRDMFD<br/> VFPJUDEEHZWETZYVGWHKKQETGFQJNCE<br/> GGWHKK?DQMCPPQZDQMMIAGPFXHQRLG<br/> TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA<br/> QZGZLECGYUXUEENJTBILBQCRTBJDFHRR<br/> YIZETKZEMVDUFKSJHKFWHKUWQLSZFTI<br/> HHDDDUVH?DWKBFUFPWNTDFIYCUQZERE<br/> EVLDFKEZMOQQJLTTUGSYQPFEUNLAVIDX<br/> FLGGTEZ?FKZBSFDQVGOGIPUFXXHHRKF<br/> FHQNTGPAECNUVPDJMQCLQUMUNEDFQ<br/> ELZZVRRGKFFVOEEXBDMVPNFQXZLGRE<br/> DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP<br/> DQUMEBEDMHDAFMJGZNUPLGEWJLLAETG</p> | <p>ABCDEFGHIJKLMNOPQRSTUVWXYZABCD<br/> AKRYPTOSABCDEFGHIJLMNQUVWXXZKRYPT<br/> BRYPTOSABCDEFGHIJLMNQUVWXXZKRYPT<br/> CYPTOSABCDEFGHIJLMNQUVWXXZKRYPTO<br/> DPTOSABCDEFGHIJLMNQUVWXXZKRYPTOS<br/> ETOSABCDEFGHIJLMNQUVWXXZKRYPTOSA<br/> FOSABCDEFGHIJLMNQUVWXXZKRYPTOSAB<br/> GSABCDEFGHIJLMNQUVWXXZKRYPTOSABC<br/> HABCDEFGHIJLMNQUVWXXZKRYPTOSABCD<br/> IBCDFGHIJLMNQUVWXXZKRYPTOSABCDE<br/> JCDFGHIJLMNQUVWXXZKRYPTOSABCDEF<br/> KDEFGHIJLMNQUVWXXZKRYPTOSABCDEFG<br/> LEFGHIJLMNQUVWXXZKRYPTOSABCDEFGH<br/> MFGHIJLMNQUVWXXZKRYPTOSABCDEFGHI</p> |
| <p>ENDYAHROHNLSRHEOCPTIOIBIDYSHNAIA<br/> CHTNREYULDSLSSLNOHSNOSMRWXMNE<br/> TPRNGATIHNRRAPESLNNELEBLPIIACAE<br/> WMTWNDITEENRAHCTENEUDRETNHAEOE<br/> TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR<br/> EIFTBRSPAMHHEWENATAMATEGYEERLB<br/> TEEFOASFIOTUETUAEOTOARMAEERTNRTI<br/> BSEDDNIAAHTTMSTEWPIEROAGRIEFEB<br/> AECTDDHILCEIHSITEGOEAOSSDDRYDLORIT<br/> RKLMLHAGTDHARDPNEOHMGFMFEUHE<br/> ECDMRIPFEIMEHNLSSTTRTVDOHW?OBKR<br/> UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO<br/> TWTQJSQSSEKZZWATJKLUDIAWINFBNYP<br/> VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR</p>  | <p>NGHIJLMNQUVWXXZKRYPTOSABCDEFGHIJL<br/> OHIJLMNQUVWXXZKRYPTOSABCDEFGHIJL<br/> PIJLMNQUVWXXZKRYPTOSABCDEFGHIJLM<br/> QJLMNQUVWXXZKRYPTOSABCDEFGHIJLMN<br/> RLMNQUVWXXZKRYPTOSABCDEFGHIJLMNQ<br/> SMNQUVWXXZKRYPTOSABCDEFGHIJLMNQU<br/> TNQUVWXXZKRYPTOSABCDEFGHIJLMNQUV<br/> UQUVWXXZKRYPTOSABCDEFGHIJLMNQUVW<br/> VUVWXXZKRYPTOSABCDEFGHIJLMNQUVWX<br/> WVWXXZKRYPTOSABCDEFGHIJLMNQUVWXZ<br/> XWXXZKRYPTOSABCDEFGHIJLMNQUVWXZK<br/> YXZKRYPTOSABCDEFGHIJLMNQUVWXZKR<br/> ZZKRYPTOSABCDEFGHIJLMNQUVWXZKRY<br/> ABCDEFGHIJKLMNOPQRSTUVWXYZABCD</p> |

K1 is highlighted in Mauve, K2 in Blue, K3 in Green and K4 in Red

The remainder of this document presents analysis relating solely to parts K1 and K4.

## Initial K4 Analysis

It is often said that frequency analysis is the strongest weapon in the cryptographer's armoury and the identification of a route into K4 has proven to be no exception to this general rule. Of all the potential ideas that were publicised in the online community the possibility of K4 being a 'one time pad' of itself is the idea that the Author believes has always carried the most weight. As a result of this conviction the majority of the initial research centred heavily on such a decryption scheme.

After multiple failed attempts and various blind detours later, it has eventually been possible to arrive at a credible decryption scheme, although this is not as grandiose a claim as it first appears for reasons that will become clear later.

The decryption scheme employs a simple technique whereby the K4 cipher text is split symmetrically and laid over a reversal of itself. In this configuration the reversed second half of the cipher text plays the role of the keyword. Using the Kryptos Tableau to decrypt yields the following results.

|             |                                                   |
|-------------|---------------------------------------------------|
| Cipher Text | RACKEUAUHUKGIDCJTZXKDGWKPFZMTTPYNBFNIWAIDULKJTAW  |
| Keyword     | ?OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSQSSEKZZ  |
| Decryption  | ?YRKDKYWCBFMXYPEGCGMYJECYJQXRPPJYGYAIMNGCQREQOTBX |

For the sake of brevity I shall refer herein to the Decryption Text given above as K4<sub>d</sub>

## Frequency & Positional Analysis For K4<sub>d</sub>

| Letter | Frequency | Positions                |
|--------|-----------|--------------------------|
| Y      | 7         | 2, 7, 14, 21, 25, 33, 35 |
| C      | 4         | 9, 18, 24, 41            |
| G      | 4         | 17, 19, 34, 40           |
| X      | 3         | 13, 28, 49               |
| P      | 3         | 15, 30, 31               |
| M      | 3         | 12, 20, 38               |
| R      | 3         | 3, 29, 43                |
| E      | 3         | 16, 23, 44               |
| J      | 3         | 22, 26, 32               |
| K      | 2         | 4, 6                     |
| B      | 2         | 10, 48                   |
| O      | 2         | 27, 46                   |
| Q      | 2         | 42, 45                   |
| I      | 1         | 37                       |
| D      | 1         | 5                        |
| W      | 1         | 8                        |
| T      | 1         | 47                       |
| A      | 1         | 36                       |

## Further Analysis Of K4<sub>d</sub>

It is immediately obvious that K4<sub>d</sub> is not readable plaintext, but a number of structural factors do stand out immediately. Of interest is the positional relationship information for the characters X and Y. These were marked as being of interest due to the Cartesian flavour of the K2 decryption keyword ABSCISSA.

If every seventh letter in K4<sub>d</sub> is skipped then it is seen that the resultant text reads off as YYYXYQX which to the eye looking for Cartesian linkages stands out immediately as a possible aspect of some underlying structural artefact.

*It is worth pointing out at this juncture that all of the discoveries that follow were not immediately obvious despite being the key to all the doors that followed. The line of enquiry that the specific Y, X and Q combination prompted were all eventual dead-ends that explored a range of different encryption schemes including morse and pseudo cursor operations.*

## Notes On Computational Analysis

It is interesting to note in hindsight that computational analysis does not offer any major help from this point in as the text starts to descend in a series of structural operations and cryptic clues that would wrong foot any potential pattern matching software. Combined with the disparity in the popularity of cryptic crosswords in America as opposed to Britain where they are pervasive it does not surprise me that there has been massive asymmetry in the time taken to find a route into K4.

## The Forensics Of Kryptos

From investigations to date on what lies beyond this point, it is also possible to venture a guess that the cipher texts of K2 and K3 were laid down very early on in the project and that K1 and K4 were produced in tandem with each other, with the wording of the K1 plaintext being significantly reworked until the desired result was achieved. The idea that K2 and K3 were already done and dusted combined with the amount of effort required to generate the desired effect for the K1 & K4 combination may well have been a factor in the 'aesthetic mistake' inherent in K2.

## Claims As To The Structural Composition Of K1/K4 And Initial Attempts To Harvest Potential Plain Texts

The following eight statements are now offered up for critique and the corresponding analysis presented in more detail in the following sections. The ninth and tenth unqualified statements should be disregarded as tenuous, but note taken that a meaningful plaintext may be compressed into 44 characters.

1. The decryption of K4 per se is: ?YRKDKYWCBFMXYPEGCGMYJECYJOXRPPJYGYAIMNGCQREQOTBX (K4<sub>d</sub>)
2. K4<sub>d</sub> interacts with both the K1 cipher and plain texts to reveal further structural information.
3. K4<sub>d</sub> is subdivided into 3 further components K4<sub>d</sub>-1, K4<sub>d</sub>-2 and K4<sub>d</sub>-3 that in their entirety comprise 44 characters.
4. Each unique section just like the four sections of the complete cipher text as a whole has its own distinct method of encryption that must be broken.
5. It is ventured that the first layer decryption of K4<sub>d</sub>-1 is LCOXWHXMAKE.
6. K4<sub>d</sub>-1, K4<sub>d</sub>-2 & K4<sub>d</sub>-3 are each DOUBLY encrypted.
7. Once the first layer of decryption is removed from each of the three parts, the final plaintext is recovered through the use of two keywords of 5 and 6 characters in length respectively. These are applied repeatedly four times against the underlying 44 character string to recover the eventual plaintext.
8. It is ventured that the first keyword in this 5 and 6 character combination is QUXQB which corresponds to the K1 cipher text for the word LIGHT.
9. It is tenuously ventured that the second keyword in plain text may either be HIOOGH where OO represents underlying characters that should remain unchanged or GNOMON which can be derived tenuously from the same original source.
10. It is conjectured that the first five letters of the K4 plain text proper are WHATI and an unqualified guess ventured that this may well correspond to a quote by Andre Malraux.

*High confidence is placed in statements 1 – 3, but evidence supporting statements 4 -8 is based solely on work undertaken against K4<sub>d</sub>-1 and is deemed speculative until further supporting evidence is found.*

## Corresponding Analysis

### Statement 1. The decryption of K4 per se is:

**?YRKDKYWCBFMXYPEGCGMYJECYJOXRPPJYGYAIMNGCQREQOTBX**

The arrival at this conclusion is irrespective of the frequency and positional analysis and is obtained solely by the ability to obtain the results that follow. If this statement is incorrect then all proceeding statements must also be a fallacy.

### Statement 2. K4<sub>d</sub> interacts with both the K1 cipher and plain texts to reveal further structural information.

If we treat the keyword PALIMPSEST used to decrypt K1 as cryptic clue, then it may indicate that K1 itself has some additional properties that have passed unnoticed to date. Palimpsest essentially means to write on top of an existing document that has since been erased, so our interest immediately centres back onto the cipher text of K1 itself. If we compare the cipher text of K1 with K4<sub>d</sub> then we start to see that a series of new structural artefacts stand out.

K4<sub>d</sub> YRKDKYWCBFMXYPEGCGMYJECYJOXRPPJYGYAIMNGCQREQOTBX

K1<sub>c</sub> EMUFPHZLRFA XY USDJKZLDKRNSHGNFIVJYQTQUXQBQVYUVLLTREVJYQTMKYRDMFD

Given the Cartesian flavour of the K2 keyword ABSCISSA , the word ORDINATE should also be close to hand in our vocabulary of potential keywords. The ABSCISSA and ORDINATE are more commonly known as X and Y. Identifying the relative positions of the XY pairing in both strings leads us to notice some other significant structural features that are outlined below.

*The following tables show the K4<sub>d</sub> text overlayed on top of the K1 Cipher Text (K1c) then the K1 Plaintext (K1p) and finally the K1 decryption Keyword PALIMPSEST cycling underneath.*

|             |            |                    |    |                    |
|-------------|------------|--------------------|----|--------------------|
| YRKDKYWCBFM | XY         | PEGCGMYJECYJOXRPP  | JY | GYAIMNGCQREQOTBX   |
| EMUFPHZLRFA | XY         | USDJKZLDKRNSHGNFIV | JY | QTQUXQBQVYUVLLTREV |
| JY          | QTMKYRDMFD |                    |    |                    |
| BETWEENSUBT | LE         | SHADINGANDTHEABSEN | CE | OFLIGHTLIESTHENUAN |
| CE          | OFIQLUSION |                    |    |                    |
| PALIMPSESTP | AL         | IMPSESTPALIMPSESTP | AL | IMPSESTPALIMPSESTP |
| AL          | IMPSESTPAL |                    |    |                    |

We can see immediately that there is a significant interaction between K4<sub>d</sub>, K1<sub>c</sub> and K1<sub>p</sub> whereby the text is broken into 3 sections in K4<sub>d</sub> and 4 sections in K4<sub>c/p</sub>. It is probably no coincidence that the Kryptos Tableau interactions at these same points all read 'AL' and it is suspected that this is a non cryptic reference implying 'Another Layer'.

If we assume that XY and JY are purely structural spacers then we are led directly to our next deduction which is that a further positional rearrangement is required in order to bring QTQUXQBQVYUVLLTRE VJY directly underneath USDJKZLDRNSHGNFI VJY. We can see that in this instance VJY now becomes the common structural spacer.

|             |    |                   |     |                    |    |            |
|-------------|----|-------------------|-----|--------------------|----|------------|
| YRKDKYWCBFM | XY | PEGCGMYJECYJOXRPP | JY  | GYAIMNGCQREQOTBX   |    |            |
| EMUFPHZLRFA | XY | USDJKZLDRNSHGNFI  | VJY |                    |    |            |
|             |    | QTQUXQBQVYUVLLTRE | VJY | QTMKYRDMFD         |    |            |
| BETWEENSUBT | LE | SHADINGANDTHEABSE | NCE | OFLIGHTLIESTHENUAN | CE | OFIQLUSION |
| PALIMPSESTP | AL | IMPSESTPALIMPSEST | PAL | IMPSESTPALIMPSESTP | AL | IMPSESTPAL |

### Statement 3. K4<sub>d</sub> is subdivided into 3 further components K4<sub>d</sub>-1, K4<sub>d</sub>-2 and K4<sub>d</sub>-3 that in their entirety comprise 44 characters

From the preceding analysis we may deduce that K4<sub>d</sub> actually gives rise to a new puzzle consisting of three distinct parts. If we remove the structural spacers then we are left with.

K4<sub>d</sub>-1 (11 characters)

|                            |             |
|----------------------------|-------------|
| K4 <sub>d</sub> Fragment 1 | YRKDKYWCBFM |
| K1 <sub>c</sub> Fragment 1 | EMUFPHZLRFA |
| K1 <sub>p</sub> Fragment 1 | BETWEENSUBT |
| K1 Keyword                 | PALIMPSESTP |

K4<sub>d</sub>-2 (17 characters)

|                            |                   |
|----------------------------|-------------------|
| K4 <sub>d</sub> Fragment 2 | PEGCGMYJECYJOXRPP |
| K1 <sub>c</sub> Fragment 2 | USDJKZLDRNSHGNFI  |
| K1 <sub>c</sub> Fragment 3 | QTQUXQBQVYUVLLTRE |
| K1 <sub>p</sub> Fragment 2 | SHADINGANDTHEABSE |
| K1 <sub>p</sub> Fragment 3 | OFLIGHTLIESTHENUA |
| K1 Keyword                 | IMPSESTPALIMPSEST |

K4<sub>d</sub>-3 (16 characters)

|                            |                  |
|----------------------------|------------------|
| K4 <sub>d</sub> Fragment 3 | GYAIMNGCQREQOTBX |
| K1 <sub>c</sub> Fragment 4 | QTMKYRDMFD       |
| K1 <sub>p</sub> Fragment 4 | OFIQLUSION       |
| K1 Keyword                 | IMPSESTPAL       |

One has to stand back at this point and momentarily forget the cryptographic aspects of Kryptos and marvel at the skill inherent in what the artist has managed to achieve here. The concept of Chinese Boxes or Russian Doll's immediately spring to mind as we suddenly find ourselves confronted by another three doors left to unlock.

**Statement 4. Each unique section just like the four sections of the complete cipher text has its own distinct method of encryption that must be broken.**

It can be seen immediately that the structures of K4<sub>d</sub>-1, K4<sub>d</sub>-2 and K4<sub>d</sub>-3 are all quite distinct from each other and are most probably encrypted using either separate Keywords or different encryption methods.

**Statement 5. It is ventured that the first layer decryption of K4<sub>d</sub>-1 is LCOXWHXMAKE**

By using K4<sub>d</sub> Fragment 1 as the cipher text and K1<sub>c</sub> Fragment 1 as the corresponding keyword to decrypt, the following plain text was extracted:

LCOXWHXMAKE.

The discovery of this plaintext was thought at the time to be significant as it might imply an entire plain text yet to be discovered that detailed a scenario such as a stand down order from the Launch Control Office of the Whitehouse. Significant research was undertaken as to the acronyms in use by the various strategic bodies at the time of the Cuban missile crisis as any resultant plain texts might be obscured by intelligence specific grammar's that appear meaningless in plain English . *Examples include ALCC, SAC & ICBM etc.*

**Statement 6. K4<sub>d</sub>-1, K4<sub>d</sub>-2 & K4<sub>d</sub>-3 are each DOUBLY encrypted.**

Initial thoughts on the decryption of K4<sub>d</sub>-1 being the true plain text are now thought to be over simplified despite its readability and it was eventually reasoned that it was subject to a secondary encryption mechanism. This conclusion was reached on the basis of working on potential decryption schemes for K4<sub>d</sub>-2 which so far have yielded no meaningful results.

**Statement 7. Once the first layer of decryption is removed from each of the three parts the final plaintext is recovered through the use of two keywords of 5 and 6 characters in length respectively. These are applied repeatedly four times against the underlying 44 character string to recover the eventual plaintext.**

As more blind alleys were reached, especially in regards to work on breaking  $K4_d-2$  it was reasoned that a secondary layer of encryption was likely to be in play.

This conclusion then started to gain some momentum after reviewing the cryptic clue embedded in K2 that was subject to personal intervention by Sanborn after parts K1-3 had been decrypted:

THIS WAS HIS LAST MESSAGE. X THIRTY-EIGHT DEGREES FIFTY-SEVEN MINUTES SIX POINT FIVE SECONDS NORTH SEVENTY-SEVEN DEGREES EIGHT MINUTES FORTY-FOUR SECONDS WEST X LAYER TWO.

If this cryptic clue wasn't taken at face value as being the coordinates of the sculpture itself then we could alternatively just write out the following number string: 38576.577844. The importance of this is that we have a five and six digit number that if repeated four times would overlay our 44 character solution exactly.

**Statement 8. It is ventured that the first keyword in this 5 and 6 character combination is QUXQB which corresponds to the K1 cipher text for LIGHT.**

Given that there was only confidence in a potential solution to  $K4_d-1$  a concerted efforts was made to ascertain the meanings of both the 5 and 6 digit numbers respectively. As  $K4_d-1$  was 11 letters long it was thought that this would be the perfect benchmark for uncovering any potential second layer of encryption.

Two more cryptic clues now came into play: The first is from the K3 plaintext.

I INSERTED THE CANDLE AND PEERED IN THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER BUT PRESENTLY DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST X  
CAN YOU SEE ANYTHING Q

The second is from the K1 plaintext.

ABSENCE OF LIGHT

The expression for light in the K1 cipher text is QUXQB. If this is metaphorically held up to the first five letters of  $K4_d-1$  and decrypted using the Kryptos Tableau then we derive a plain text of WHATI.

Given that the key to solving the overall puzzle most probably lies with  $K4_d-1$ . This cryptic clue left in K3 now takes on some additional meaning.

I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER

It began to become apparent that the journey undertaken to this point was very similar to the opening text of K3. It was reasoned that the structural operations taken out on K1 as a result of  $K4_d$  had indeed been like breaking off a piece of door way. It leads to the overall question of whether K3 is actually a metaphorical description of how the K4 plain text is derived from start to finish.

**Statement 9. It is tenuously ventured that the second keyword in plain text may either be HIOOGH where OO represents underlying characters that should remain unchanged or GNOMON which can be derived tenuously from the same original source.**

This statement has been grouped along with statement ten as being extremely tenuous. Regular solvers of daily cryptic crosswords may sympathise with its merits, whilst others may feel ill at ease with the route necessary in order to extract the end result. If we return to our original number string from statement 7, then we have our two numbers consisting of five and six digits a piece, 38576 and 577844. If LIGHT is indeed the correct secondary decryption keyword then we can assume that there is a relationship between 38576 and the plain text LIGHT. We know that we can only be dealing with numbers one through to eight in regards to both numbers so the letters of LIGHT can be laid out in the following positions:

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|   |   | L |   | G |   | T | H |
|   |   |   |   |   |   |   | I |

It is immediately obvious that we have a table that is missing three letters. Various words were constructed that would fill the missing gaps with an example being THINKLOG, which was thought to sound promising as it would seem to compel us to 'think logically', but it was then noted almost instantly that a major clue in the sculpture itself may offer some cryptic assistance here. The sculpture itself stands next to a large piece of petrified wood which thinking cryptically may also be thought of as a 'thick log'.

If we now substitute in the missing letters of THICKLOG we arrive at:

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| C | K | L | O | G | T | H | I |

If we now read off the plain text for the number value 577844 then we arrive at: GHHIOO.

Shifting the letters to the left twice gives us HIOOGH. In tandem with the word LIGHT it would seem to suggest HIGH LIGHT and one would assume that OO may imply a scheme whereby the underlying letters uncovered in the first decryption remain unchanged (much like the action of light from a flickering candle).

The feeling cannot be shaken that GHHIOO may alternatively be alluding to the word GNOMON which has a meaning in its original Greek of 'that which reveals' which would fit sympathetically with the use of the word LIGHT. This would seem to imply that there is something spatial about the final decryption that may only be seen at the site of the sculpture at a certain hour of day as many have already suggested.

The important information that hasn't been imparted here is how HIOOGH or GNOMON itself may be encrypted for use as a Keyword. It is presently assumed that PALIMPSEST must still be the keyword in play, but at present no evidence has been found to support this conjecture.

**Claim 10. It is conjectured that the first five letters of the K4 plaintext proper are WHATI and an unqualified guess ventured that this may well correspond to a quote by Andre Malraux.**

If the assertion that the second keyword utilises a two character space that preserves the underlying layer 1 decryption is correct, then the final doubly decrypted plain text of K4<sub>d</sub>-1 would read as WHATI\_\_MA\_\_. Any number of combinations would be plausible for filling the complete 44 letters of the required plain text such as WHAT IS IMAGINATION etc, but the information imparted publicly by the ex CIA Director in relation to K4 implied that the complete answer was 'philosophical and obscure'. What is effectively a social hack led to a brief search for potential quotes that would fit such criteria.

The quote 'WHAT IS A MAN? A MISERABLE LITTLE PILE OF SECRETS' attributed to Andre Malraux would appear a fitting conclusion given the nature of the work undertaken at the CIA. Malraux would also fit as a potential source known to Sanborn as he wrote widely on a number of subjects including art and sculpture.

Compressed into 44 characters it is conjectured that such a plain text would read something like:

WHATISAMANXAXMISERABLELITTLEXPILEXOFXSECRETS

Should anyone search the internet in pursuit of a source for this quote, then they will no doubt come across multiple references from its use in a video game published by KONAMI in the 1990's. In the time since the sculpture was installed the game became a hit, and its obscure dialogues passed into the vocabulary of teen sub culture. As a result there are multiple sites on the internet where T-shirts or mugs emblazoned with the same wording may be purchased. Given that the final plain text may not be revealed for some time to come the Author is happy to settle on this conclusion as he notes with a wry smile that teenagers all over the western world may have been walking around the streets keeping the secret in plain sight for most of the last decade.

## **Closing Remarks**

It is hoped that the newly revealed inner structure of K4/K1 outlined in statements one to three will offer an exciting opportunity to break the K4 impasse of the last decade. It is also hoped that there is some meaningful signal to be extracted from within the noise of statements four to ten.

If anyone would like to offer a critique, corrections or suggest related collaborations then feel free to email [K4@dataluminescence.co.uk](mailto:K4@dataluminescence.co.uk) where I will endeavour to try and respond.

## Feedback

Useful feedback has already started to filter back from the Kryptos forums, so an effort will be made to try and keep a log of how the basic premise of statements 1 -3 progresses, unravels or mutates.

Monday 11<sup>th</sup> October 2010

Progression. Feedback on changes necessary to background information.

Tuesday 12<sup>th</sup> October 2010

Unravelling. It is pointed out that JS has recently indicated that K4 plain text is approximately 100 characters long.

Thursday 14<sup>th</sup> October 2010

Major Coincidence. Whilst digging up some information about the artist Kit Williams I stumble across a picture of the Sanborn copper screen installation in Stamford CT. A shiver goes down my back as I realise that I must have walked past that sculpture every day when I was visiting the company offices in the US. In fact in one of the shots our old company offices are actually visible.

Progression. Kryptos Group Member Dan points me at some work on his compass rose theory. I strip the theory back to its original premise and redo the rose using the original cipher text. Compare against K4<sub>d</sub> to see if anything stands out.

Progression. Studying group member Dan's Compass Rose theory suddenly highlights the fact that the two misspelt letters are A and U. I point out that this may be an allusion to the golden section or phi. Transpires that there is a binary sequence called the Golden String closely related to Fibonacci numbers that has a self similar repeating element of 101101.

Tuesday 19<sup>th</sup> October 2010

Progression. Two things strike me today. The first is that the original document above wrongly stated the order in which the decryption is done. The OBKR part of the string should have read as being the Keyword used to decipher ACKEUA etc. Document corrected.

The second is that if we don't terminate halfway and incorporate the entire K4 string then we end up with the following relationship.

CT  
ACKEUAUHUHGIDCJTZXKDGWKPFZMTTPYBNFNIWAIDULKJTAWZZKESQJSQWTOSKGNRPQQVRLFWBFFILOSBLUHGOXOURK  
BOR

KW  
OBKRUXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJSSEKZZWATJKLUDIAWINFBNYPVTMTZFPKWGDZXTJCDIGKUHUAUEK  
CAR

PT  
YRKDKYWCBFMYPEGCGMYJECYJXRPPJYGYAIMNGCQREQOTBXMYMVUSIZSLGABENXGXDWWZYUDXLIDXBGLGIWXYBHMLPXKJK  
ZXX

Of interest is the resulting XY combos that are present in the string similar to the first one we used to separate out  $K4_d-1$ .